

Akhilesh Varute

Software Engineer | AI Infrastructure & Cloud Platforms

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SUMMARY

Software Engineer with 2 years building production AI and cloud infrastructure on AWS. Primary engineer on aicloudinsights.ai at GPUonCLOUD — a live multi-tenant SaaS using Amazon Bedrock for natural-language AWS infrastructure monitoring. Strong in Node.js, TypeScript, AWS (Bedrock, Lambda, STS, CloudWatch), Express.js, REST APIs, and MongoDB. AWS Certified Developer – Associate.

PROFESSIONAL EXPERIENCE

Expert Cloud Consulting aka GPUonCLOUD, Software Developer 06/2024 – Present | Pune

- Primary engineer on aicloudinsights.ai — built and shipped every feature of the live multi-tenant SaaS end-to-end. Drives architectural decisions and reviews PRs across a 3-person team using GitHub Projects.
- Built a Bedrock-powered (Claude) conversational interface using function calling / tool use — backend defines AWS tool schemas (Cost Explorer, Security Hub, GuardDuty, CloudWatch, IAM); Claude parses user intent in plain English, selects the right tool, and the system executes the API call and returns real-time answers across multiple connected accounts.
- Designed cross-account architecture using IAM role assumption via STS — strict per-tenant data isolation across 8+ AWS services per account, no shared credentials.
- Built a parallel cost optimisation engine analysing 8 AWS services simultaneously using CloudWatch + Pricing API; findings auto-ranked by dollar impact (critical / high / medium / low).
- JWT-based auth + RBAC stored in MongoDB with per-user, per-account visibility. CI/CD via CodePipeline + CodeBuild; CloudWatch dashboards for production monitoring.
- Previously: Kubernetes-native infrastructure provisioning engine (Crossplane + Upbound + Bedrock) — plain-English AWS resource description → automatic CRD provisioning, GitOps-style with drift detection. Internal R&D prototype; deprioritised before broader rollout due to funding reallocation.
- Mentored 2 junior engineers through code reviews, architecture standards, and technical guidance.

EDUCATION

Post Graduate Diploma in Advanced Computing (PG-DAC), Sunbeam Infotech 2024

Bachelor of Technology, DKTE Institute of Technology 2022

TECHNICAL FOCUS AREAS

Cloud & AI	Infrastructure	Backend	Auth & Data
AWS Bedrock, Lambda, API Gateway, STS, CloudWatch, Cost Explorer, Security Hub, S3, EC2	Docker, Kubernetes, CI/CD, CodePipeline, GitHub Actions	Node.js, TypeScript, Express.js, Fastify, REST APIs, Microservices	JWT, RBAC, Multi-tenant architecture, MongoDB, Redis
Frontend	AI / GenAI	AI Dev Tools	
React.js, JavaScript	Amazon Bedrock, LLMs, Function calling / Tool use, Prompt Engineering, NLP	Claude Code, GitHub Copilot, Claude (Anthropic)	

CERTIFICATIONS

- AWS Certified Developer – Associate (DVA-C02)

PROJECTS

AI Cloud Insights Platform, Node.js, TypeScript, Express.js, Lambda, Bedrock (Claude), STS, CloudWatch, MongoDB, RBAC, Docker

- Live multi-tenant SaaS — enterprises connect AWS accounts and query costs, security posture, and resource usage in plain English. Cross-account STS; cost engine ranks 8 services by dollar impact.
- JWT + RBAC stored in MongoDB with per-user, per-account visibility across the multi-tenant deployment.

Cloud Infrastructure Automation Platform, Node.js, Fastify, Bedrock, Crossplane, Kubernetes, WebSocket, SSM, Redis, MongoDB

- Plain-English → Bedrock intent parsing → Crossplane CRD provisioning, GitOps-style with drift detection. WebSocket in-browser SSH (xterm.js + SSM). Internal R&D prototype; deprioritised before broader rollout due to funding reallocation.